Study Material Practical 32

If matrix **A** is of size m × n, then its transpose $\mathbf{A}^T$ is of size n × m.

Transpose means:

A[i][j] → A[j][i]

If a matrix **A** has size **m × n**, then its transpose $\mathbf{A}^T$ will have size **n × m**.

That means **rows become columns and columns become rows**.

Example to Understand the Procedure

Original Matrix (2 × 3)

1 2 3
4 5 6

Transpose (3 × 2)

1 4
2 5
3 6

Procedure / Algorithm for Transpose of a Matrix

Step 1: Start the program.

Step 2: Read the number of rows (m) and columns (n).

Step 3: Declare a matrix of size m × n.

Step 4: Input elements into the matrix.

Step 5: Declare another matrix of size n × m (for transpose).

Step 6: Use nested loops:

- Outer loop for rows (i = 0 to m-1)
- Inner loop for columns (j = 0 to n-1)

- Assign:

transpose[j][i] = matrix[i][j];

Step 7: Display the original matrix.

Step 8: Display the transpose matrix.

Step 9: Stop the program.